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ODD END SEMESTER EXAMINATION, DECEMBER-2025

Name of the Course: B.Tech. CSE

Semester: III

Name of the Paper: Probability and Random Processes

Paper Code: TCS-344

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1.	(20 Marks)	
(a)	A factory purchases components from three vendors V_1, V_2, V_3. They supply 50%, 30%, and 20% of the total components respectively. The defect rates for the vendors are 2%, 3%, and 4% respectively. If a component chosen at random is found to be defective: (i) Find the probability that it was supplied by vendor V_2. (ii) Find which vendor contributes the highest proportion of defective items overall.	CO1
(b)	The lifetime (in years) of a machine component follows an exponential distribution with a mean of 5 years. Find the probability that a randomly chosen component will last: (i) Less than 3 years. (ii) Between 2 and 6 years, (iii) More than 8 years.	
(c)	If X follows a normal distribution with $\mu = 50, \sigma = 5$, find (i) $P(45 < X < 55)$, (ii) $P(X > 60)$, (iii) $P(X < 40)$. Take $\phi(1) = 0.8413$ and $\phi(2) = 0.9772$	
Q2.	(20 Marks)	
(a)	Let the joint probability density function of X and Y be $f(x, y) = 8xy, 0 < x < y < 1.$ Find:	CO2

	(i) $P(X < 0.5, Y > 0.5)$ (ii) Check whether X and Y are independent.	
(b)	The random variable X has PDF $f(x) = 2x, 0 < x < 1$. Find the PDF of $Y = X^2$ using transformation of variables. Also compute $E[Y]$ and $Var(Y)$.	
(c)	The monthly salary of employees in a company is normally distributed with a mean of ₹50,000 and a standard deviation of ₹8,000. A random sample of 64 employees is selected. Find: (i) The probability that the sample mean salary exceeds ₹51,500. (ii) The probability that the sample mean lies between ₹48,500 and ₹51,000. (iii) Determine the sample size required if the standard error of the mean is to be less than ₹1,000.	
Q3.	(20 Marks)	
(a)	For a random process $X(t) = 10 + 3 \cos(2\pi 100t + \theta)$, where θ is uniformly distributed over $[0, 2\pi]$, find: (i) the mean (ii) autocorrelation function. (iii) determine whether the process is wide-sense stationary.	CO3
(b)	A process has autocorrelation $R_{XX}(\tau) = 4\cos(2\tau)e^{- \tau }$. Compute the average power and power spectral density of the process.	
(c)	The power spectral density of a process is $S_{XX}(\omega) = 2$ for $ \omega < 4$, and 0 otherwise. Find the autocorrelation function $R_{XX}(\tau)$.	
Q4.	(20 Marks)	
(a)	Customers arrive at a service counter according to a Poisson process with an average rate of 4 customers per 5 minutes. Find: (i) The probability that exactly 3 customers arrive in a 5-minute interval. (ii) The probability that at least one customer arrives in the first 2 minutes. (iii) The probability that no customer arrives in a 30-second period.	CO4
(b)	A random process $X(t) = A \cos(2\pi t + \theta)$, where A is constant and θ is uniformly distributed over $[0, 2\pi]$. Show that $X(t)$ is a wide-sense stationary process.	
(c)	A system can be in states S_1, S_2 or S_3 with the transition probability matrix $P = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.5 & 0.4 & 0.1 \\ 0.2 & 0.4 & 0.4 \end{bmatrix}$ Find the steady-state probabilities for each state.	

Q5.	(20 Marks)	
(a)	An input $x(t) = e^{-t}u(t)$ is applied to an LTI system whose impulse response is $h(t) = e^{-2t}u(t)$. Determine: (a) The output $y(t)$ using convolution. (b) The auto-correlation function $R_y(\tau)$ of the deterministic output signal.	
(b)	A stationary random process $X(t)$ with power spectral density $S_X(\omega) = \frac{10}{\omega^2 + 9}$ is applied to a linear system with transfer function $H(\omega) = \frac{1}{j\omega + 2}.$ Find the output power spectral density $S_Y(\omega)$ and the average power of the output process.	CO5
(c)	A random process $X(t)$ is input to an LTI system with impulse response $h(t) = e^{-t}u(t)$. Given that the auto-correlation function of $X(t)$ is $R_X(\tau) = e^{- \tau },$ find the cross-correlation function $R_{XY}(\tau)$ between input and output.	

